

(a) Move, don't regenerate



frozen DINOv2 patches $\mathbf{Z}_0$
+ future actions $a_1, \dots, a_{10}$
→ forecast $\mathbf{Z}_1, \dots, \mathbf{Z}_{10}$

Existing latent WMs regenerate every patch

DINO-WM, V-JEPA 2-AC, AR: re-predict all, feed back



× background drifts × moving parts blur × errors compound

ShiftWM moves what it has already seen



$$\hat{\mathbf{Z}}_k = (1-g)\mathbf{Z}_0 \text{ keep} \\ + g\mathbf{T}_k \text{ move} \\ + \mathbf{r}_k \text{ correct}$$

zoom on arm: green = gate g , arrows = transport $\mathbf{T}_{10}$

✓ every k from measured $\mathbf{Z}_0$ in parallel: no feedback

✓ plug-in head for published world models

(b) One held-out DROID window

true $t+10$ (decoded)



Direct



AR-TF (DINO-WM-style)



ShiftWM (ours)



AR (recursive)



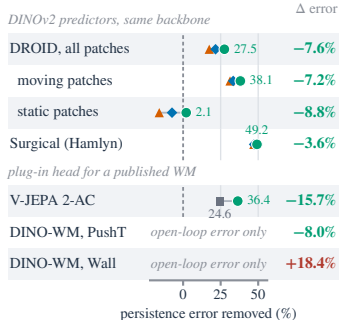
Persistence (copy t)



Each tile: that model's own forecast, one shared decoder.
Corner: feature error at $t+10$ (lower is better).

(c) Held-out results

▲ AR ◆ Direct ■ base WM ● ShiftWM



ShiftWM error below Direct in 100/100
horizon × motion-decile cells (DROID)